ECEN 474/704: Lab 02

Layout Design, Simulation and Verification in Cadence

Sheena Selvakumar
UIN: 537000654
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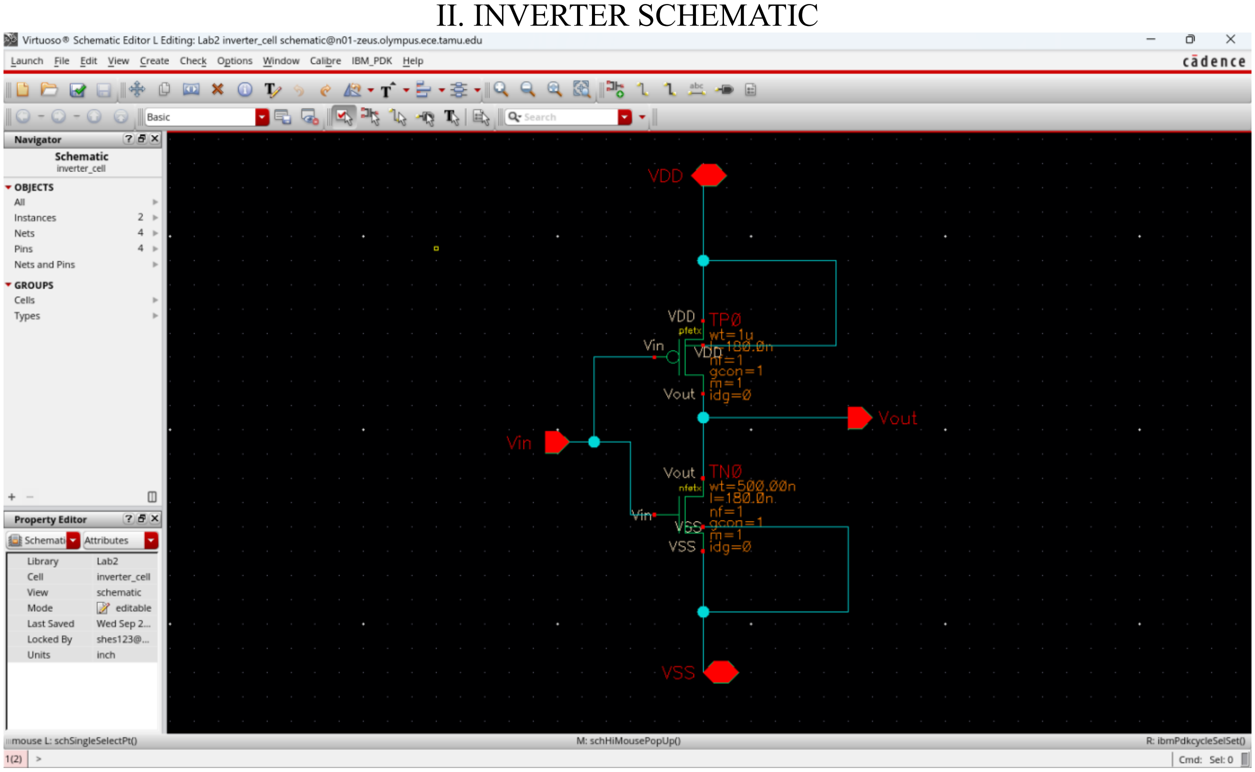
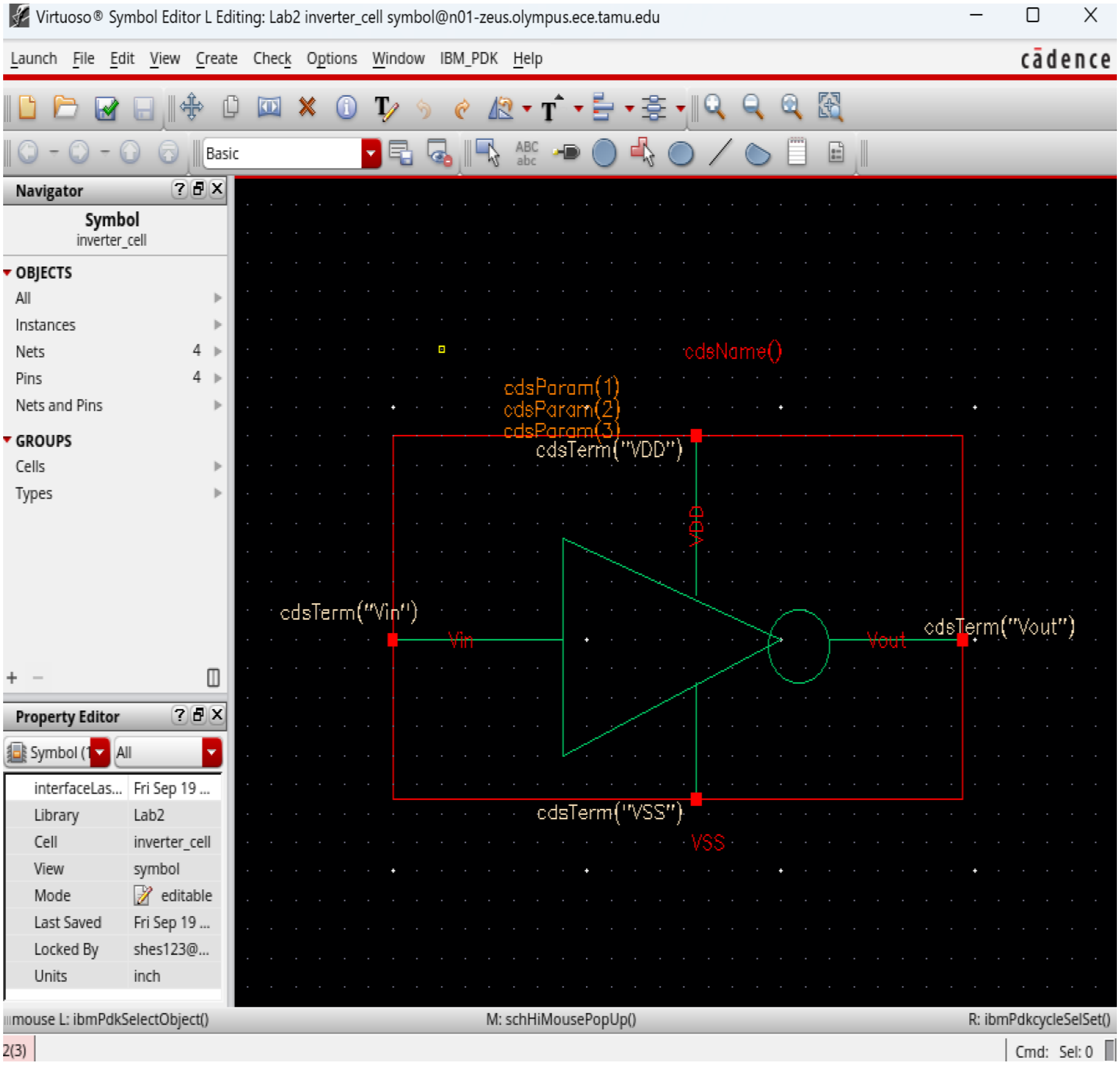
I. DISCUSSION OF THE LAB AND RESULTS

In this lab, I worked on designing, simulating, and verifying an inverter at both the schematic and layout levels. The initial schematic simulations showed the expected switching behavior and confirmed that the design worked as intended. Once the layout was created and checked with DRC and LVS, I ran the post-layout simulations. The results were very similar to the pre-layout ones, with only small differences in delay and output transitions because of the extra parasitics introduced by the layout. These changes were expected and are a good reminder that real physical designs will never behave exactly like their ideal schematics. Overall, this lab helped me see the importance of layout verification and gave me a better understanding of how parasitic effects influence circuit performance while still keeping it within acceptable limits.

II. INVERTER SCHEMATIC



Symbol creation:


Virtuoso® Symbol Editor L Editing: Lab2 inverter_cell symbol@n01-zeus.olympus.ece.tamu.edu
Launch File Edit View Create Check Options Window IBM_PDK Help
cādence
Basic
Navigator
Symbol
inverter_cell
OBJECTS
All
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Nets 4
Pins 4
Nets and Pins
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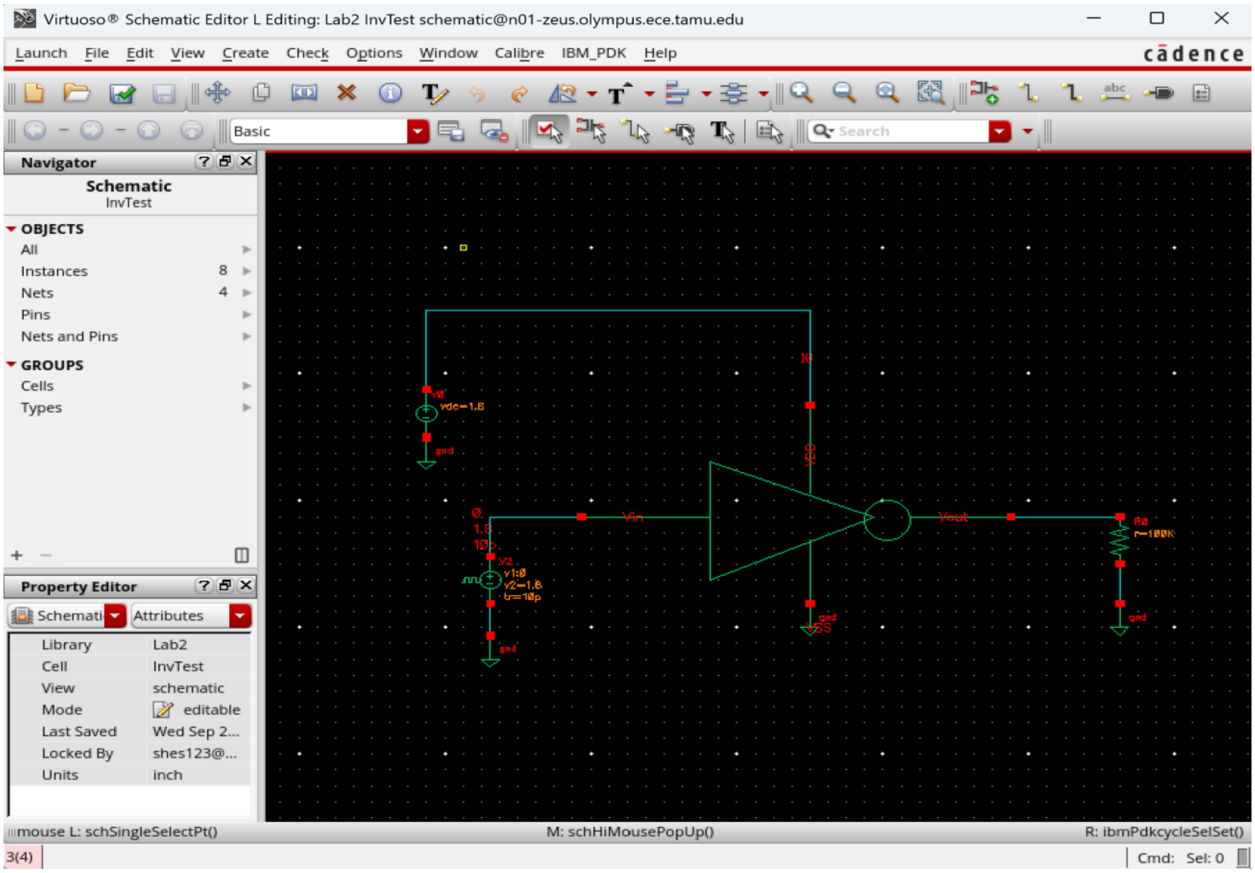
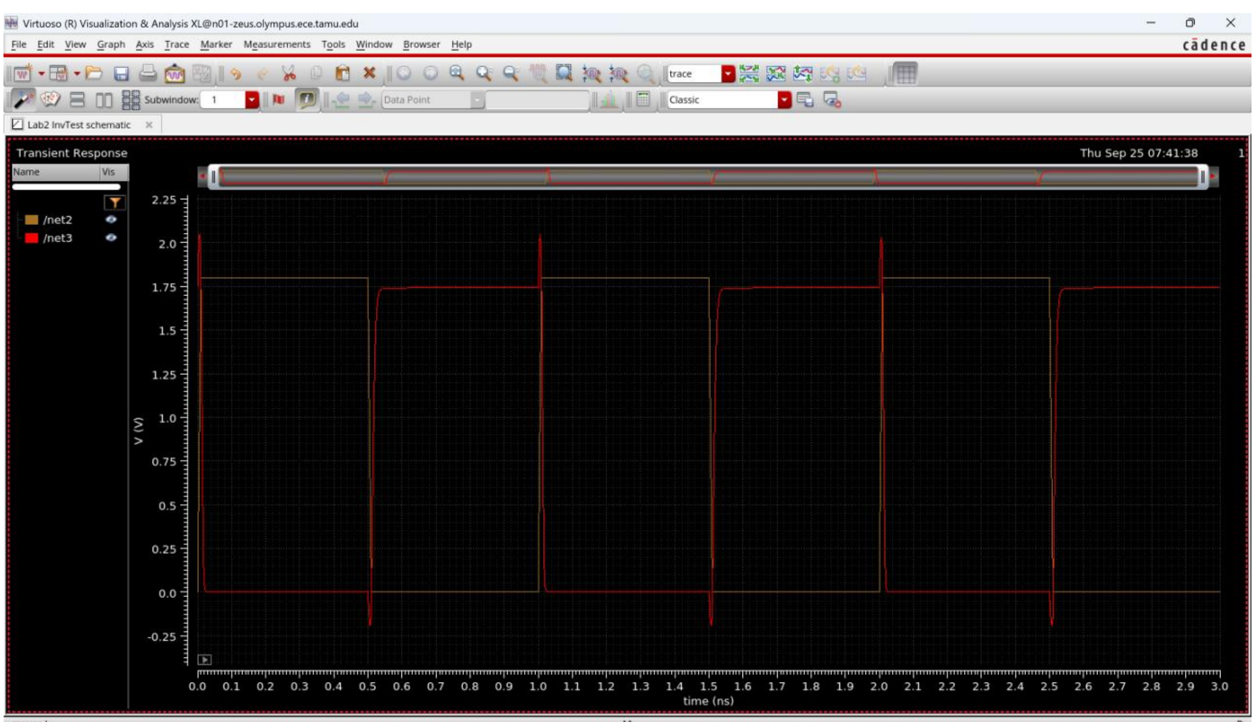
Cells
Types
Property Editor
Symbol (1
All
interfaceLas... Fri Sep 19 ...
Library Lab2
Cell inverter_cell
View symbol
Mode editable
Last Saved Fri Sep 19 ...
Locked By shes123@...
Units inch
cdsName()
cdsParam(1)
cdsParam(2)
cdsParam(3)
cdsTerm("VDD")
VDD
cdsTerm("Vin")
Vin
Vout
cdsTerm("Vout")
cdsTerm("VSS")
VSS
mouse L: ibmPdkSelectObject()
M: schHiMousePopUp()
R: ibmPdkcycleSelSet()
2(3)
Cmd: Sel: 0

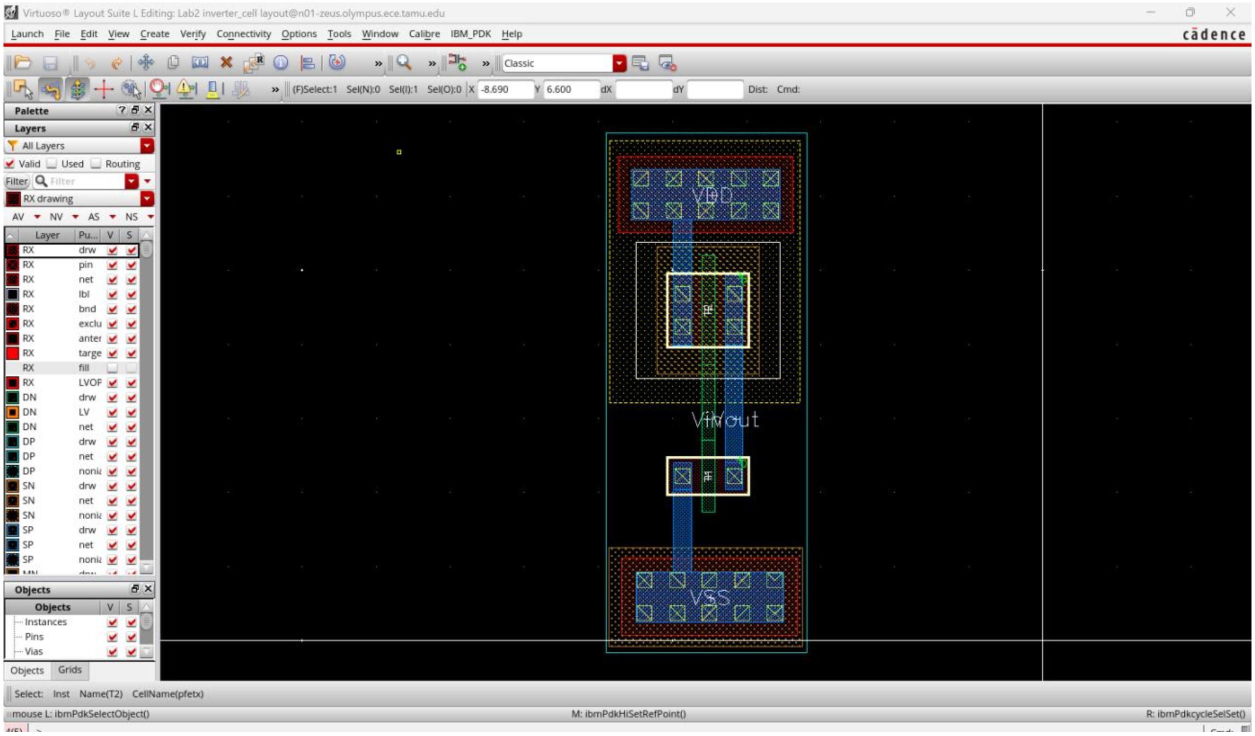
III. TEST BENCH SCHEMATIC



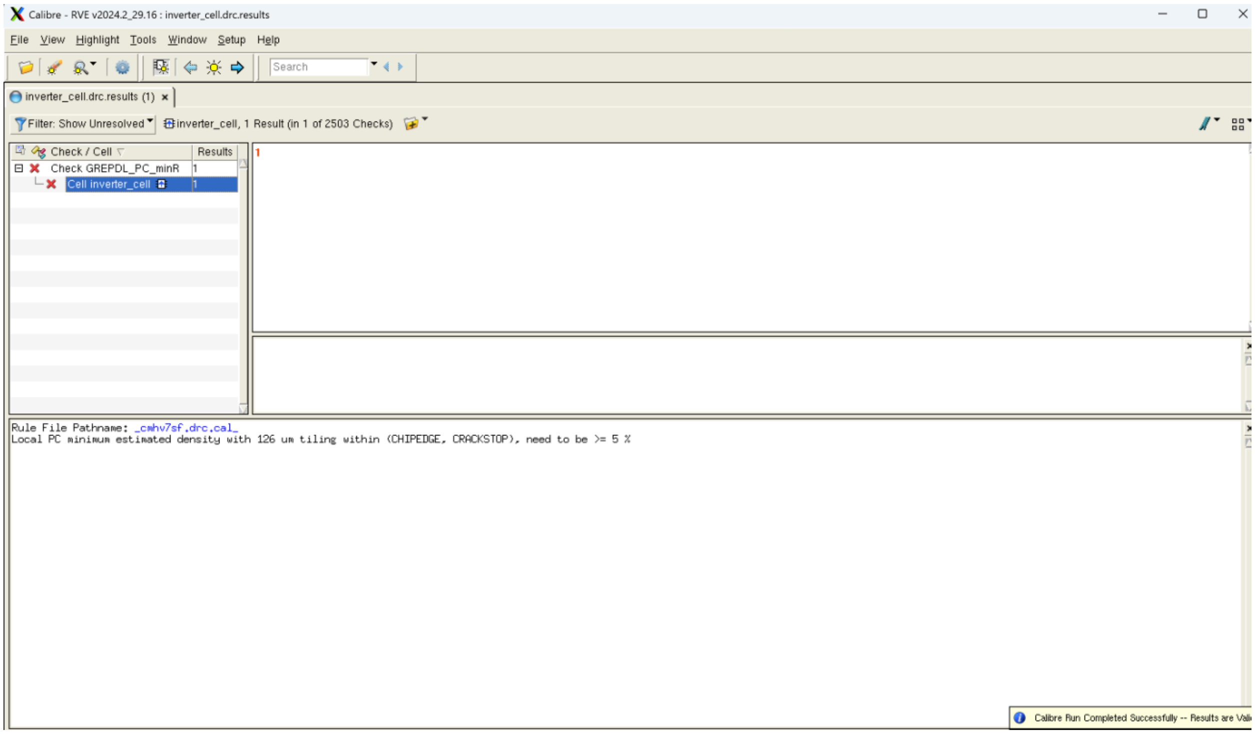
IV. TRANSIENT SIMULATION OUTPUT PLOT SHOWING VOUT AND VIN



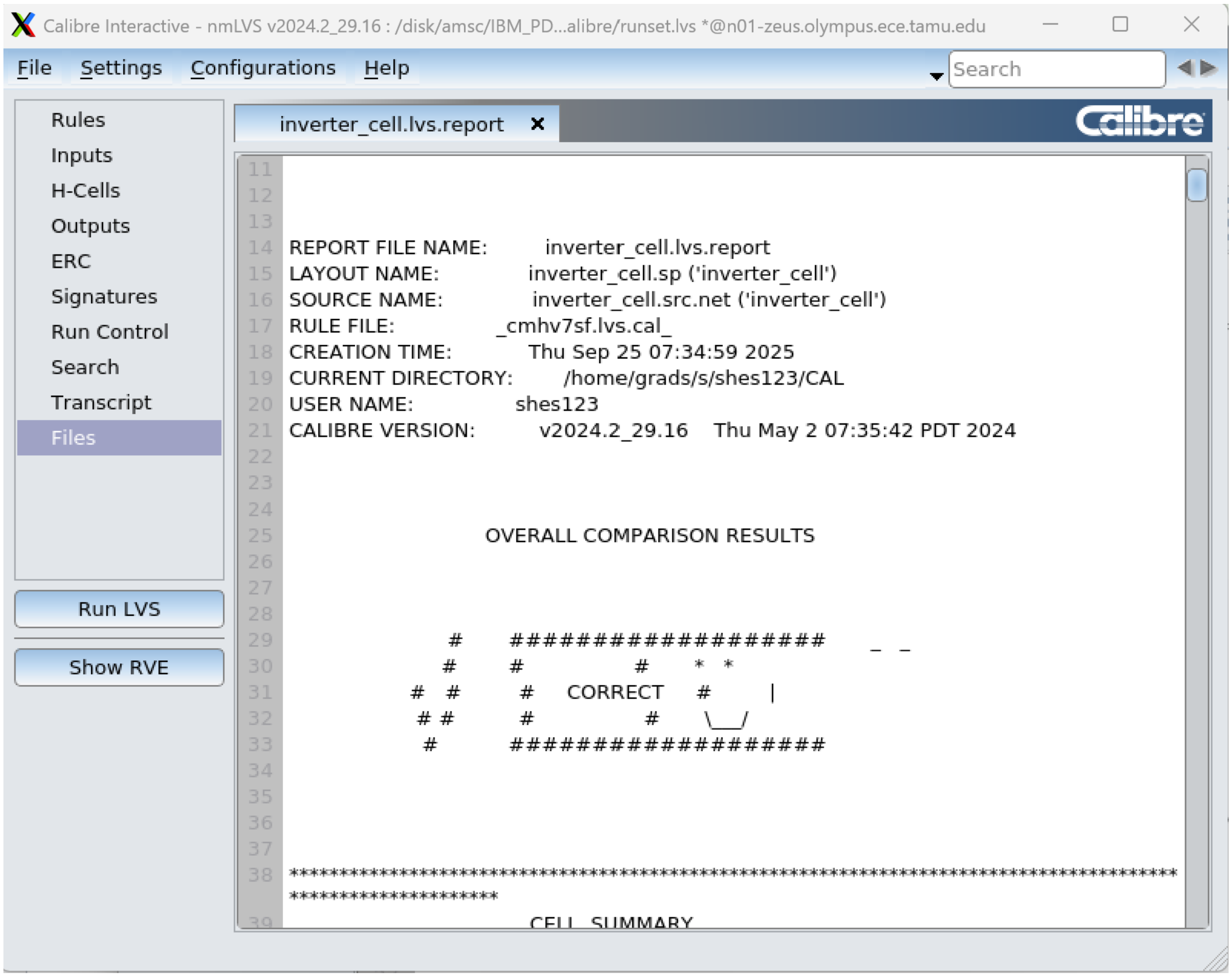
V. INVERTER LAYOUT

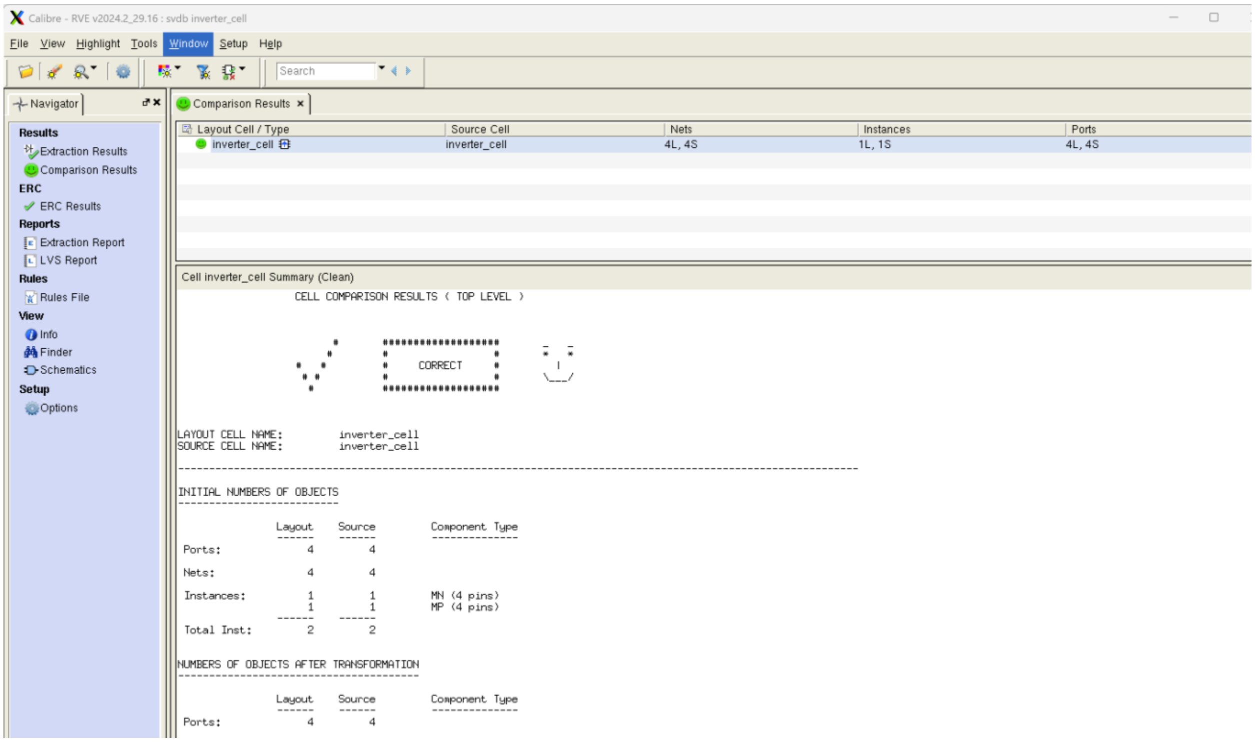


VI. DRC RVE WINDOW

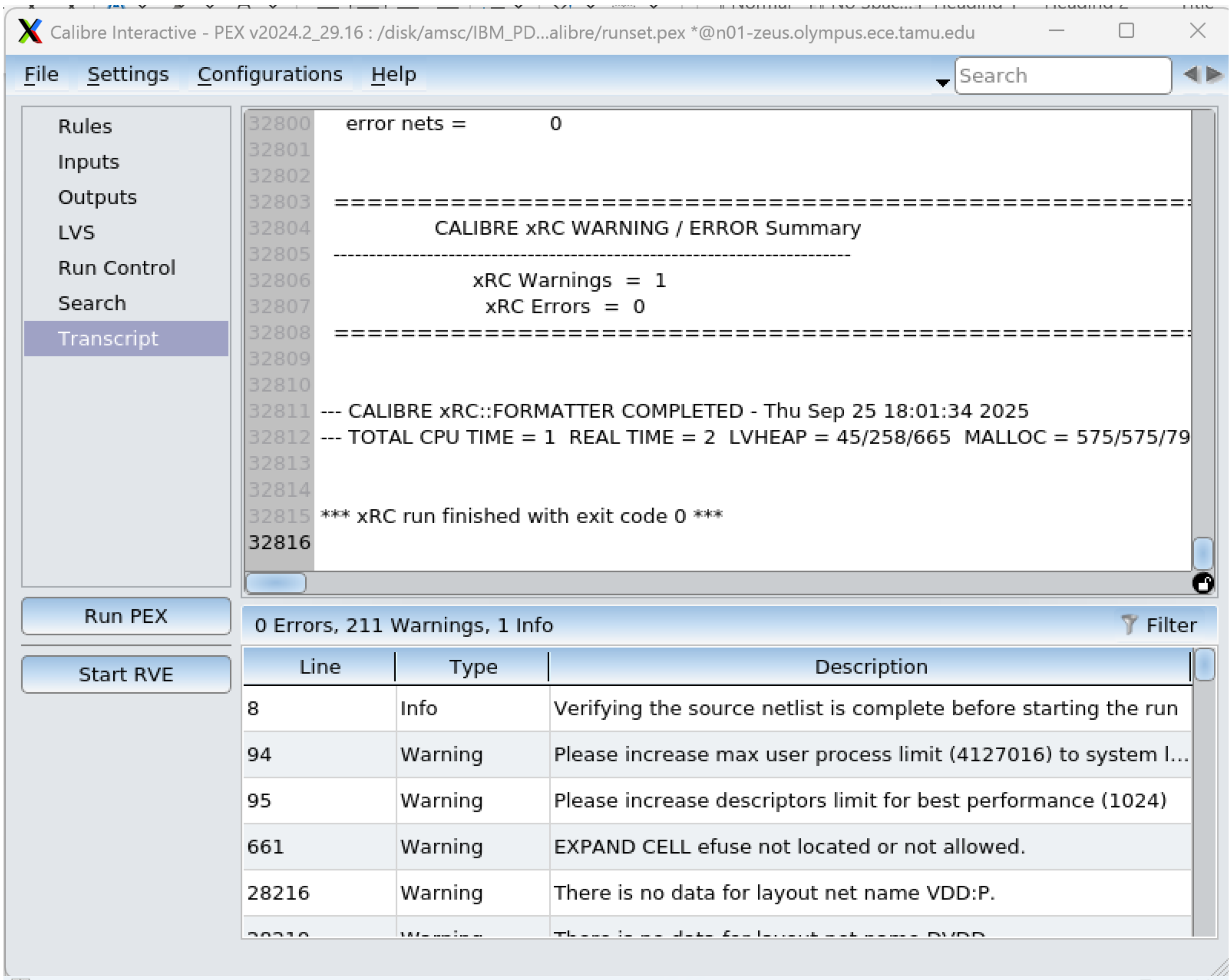


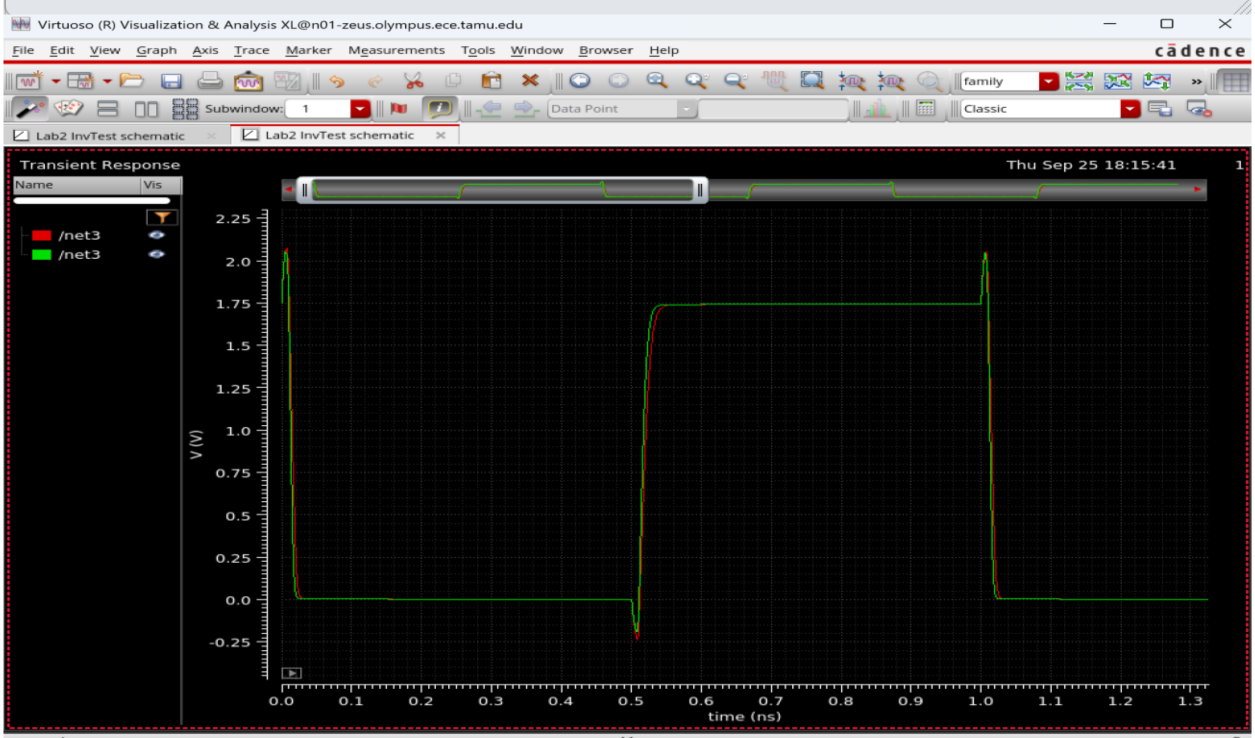
VII. LVS OUTPUT FILE


Calibre Interactive - nmLVS v2024.2_29.16 : /disk/amsc/IBM_PD...alibre/runset.lvs *@n01-zeus.olympus.ece.tamu.edu
File
Settings
Configurations
Help
Search
Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files
Run LVS
Show RVE
inverter_cell.lvs.report
REPORT FILE NAME: inverter_cell.lvs.report
LAYOUT NAME: inverter_cell.sp ('inverter_cell')
SOURCE NAME: inverter_cell.src.net ('inverter_cell')
RULE FILE: _cmhv7sf.lvs.cal_
CREATION TIME: Thu Sep 25 07:34:59 2025
CURRENT DIRECTORY: /home/grads/s/shes123/CAL
USER NAME: shes123
CALIBRE VERSION: v2024.2_29.16 Thu May 2 07:35:42 PDT 2024
OVERALL COMPARISON RESULTS
CORRECT
CELL SUMMARY



VIII. PLOT OF THE POST LAYOUT SIMULATION


Calibre Interactive - PEX v2024.2_29.16 : /disk/amsc/IBM_PD...alibre/runset.pex *@n01-zeus.olympus.ece.tamu.edu
File Settings Configurations Help
Search
Rules
Inputs
Outputs
LVS
Run Control
Search
Transcript
32800 error nets = 0
32801
32802
32803 ==
32804 CALIBRE xRC WARNING / ERROR Summary
32805 --
32806 xRC Warnings = 1
32807 xRC Errors = 0
32808 ==
32809
32810
32811 --- CALIBRE xRC::FORMATTER COMPLETED - Thu Sep 25 18:01:34 2025
32812 --- TOTAL CPU TIME = 1 REAL TIME = 2 LVHEAP = 45/258/665 MALLOC = 575/575/79
32813
32814
32815 *** xRC run finished with exit code 0 ***
32816
Run PEX
Start RVE
0 Errors, 211 Warnings, 1 Info
Filter
Line | Type | Description
8 | Info | Verifying the source netlist is complete before starting the run
94 | Warning | Please increase max user process limit (4127016) to system l...
95 | Warning | Please increase descriptors limit for best performance (1024)
661 | Warning | EXPAND CELL efuse not located or not allowed.
28216 | Warning | There is no data for layout net name VDD:P.


Virtuoso (R) Visualization & Analysis XL@n01-zeus.olympus.ece.tamu.edu
File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help
cādence
family
Subwindow: 1
Data Point
Classic
Lab2 InvTest schematic
Lab2 InvTest schematic
Transient Response
Thu Sep 25 18:15:41
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/net3
V (V)
2.25
2.0
1.75
1.5
1.25
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0.25
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-0.25
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0 1.1 1.2 1.3
time (ns)
mouse L:
M:
R: